

Orbital Stellar

The 20-Year Plan — from a Stellar event engine to the programmable runtime of Stellar's regulated-money economy

PREPARED

April 22, 2026

CLASSIFICATION

Internal — Strategic
Planning

HORIZON

20 Years (2026–2046)

AUTHOR

Founding Team

VERSION

2.0 · Replaces v1 (Apr 14)

SCOPE

Thesis · Product · Go-to-
Market

Foundation

The north star (one sentence)

Orbital is Stellar's programmable runtime — the reason a developer uses Stellar at all, and the reason a regulated money operator trusts Stellar enough to run production on it.

The buyer hierarchy

1. **Primary:** Regulated money movement — anchors, fintechs, banks, CBDCs, remittance operators.
2. **Secondary:** AI agents and their developers (x402, agent-SDK consumers).
3. **Tertiary:** Independent Stellar developers (top-of-funnel, not the revenue).

This ordering is irreversible after Month 18. Every hire, pitch, and pricing decision flows from it.

The three-tier architecture

Tier	License	Audience	Pricing	Role
Orbital Core	MIT	Everyone	Free forever	Protocol + SDK. Adoption funnel.
Orbital Cloud	Closed, operated	Startups, indie devs	\$0 / \$29 / \$149 / \$499 per month	Hosted operation of Core. Developer revenue.
Orbital Rails	Closed + certified	Fintechs, anchors, CBDCs	\$50K–\$500K / yr	Operation + certifications + ZK trust. Enterprise revenue.

Rule for the next 20 years: every feature Orbital ever builds has two halves — an OSS protocol/SDK/reference and a closed hosted operation. They grow together, never alone.

ERA I · MONTHS 0–18

Foundation & Fundability

Ship the engine. Earn the grants. Plant the flag.

Mission: ship a production-capable OSS engine; stand up minimal Cloud; earn grant + seed funding; plant the ecosystem flag.

OPEN SOURCE CORE DELIVERABLES

- PostgreSQL persistence adapter for the webhook registry
- Full Stellar operation taxonomy: `create_account` , `path_payment` , `change_trust` , `manage_offer` , `invoke_host_function`
- Soroban event decoder (ABI registry client)
- Replay store with pluggable backend (local disk, S3)
- Dead-letter queue + CLI replay tool
- HA mode (leader election via Redis/etcd)
- Prometheus / OpenTelemetry exporters
- React hooks library (`@orbital/pulse-notify`) at v1.0
- Reference Express server as a shippable Docker image
- v1.0.0 with documented stability pledge

CLOSED CLOUD MVP

- Customer account system (signup, login, team management)
- API-key issuance and scoped permissions
- Stripe billing integration
- Minimal web dashboard (view events, configure webhooks, generate keys)
- Single-region hosted Core deployment, 99.9% SLA
- Basic admin tooling

BUSINESS FUNDRAISING & ECOSYSTEM

- SDF Developer Grant application — frame as "Stellar's programmable runtime"
- Stellar Community Fund submission in parallel
- Angel round target: \$250K–\$750K from Stellar-adjacent investors
- **First SEP draft** — standardized Soroban event schema or reactor contract spec (the credibility move)

- 3 tutorials/month; 3 boilerplates: `orbital-next-starter` , `orbital-express-starter` , `orbital-anchor-starter`

SUCCESS CRITERIA — MONTH 18

≥500

NPM DOWNLOADS / WEEK

≥\$400K

EXTERNAL FUNDING

≥10

PUBLIC PRODUCTION APPS

1 SEP

ACCEPTED OR UNDER REVIEW

≥5

PAYING CLOUD CUSTOMERS

v1.0.0

CORE PUBLISHED W/ STABILITY PLEDGE

Runtime Ownership

Own the SDK surface. Scale Cloud. Begin Rails.

Mission: own the Stellar developer SDK surface; scale Cloud; begin Rails; author the standards competitors must follow.

OPEN SOURCE CORE + SDK ECOSYSTEM

- `@orbital/hooks` — complete data-hook library (`useAccount` , `useBalance` , `useTransaction` , `useOrderBook`)
- `@orbital/payments` — transaction primitives (send, receive, path payment, payroll batch)
- `@orbital/auth` — embedded wallets via WebAuthn/passkeys, fee sponsorship, WalletConnect
- `@orbital/analytics` — client library + event-volume reference dashboards
- **Soroban ABI Registry as an open standard** — schema, verification spec, client library. Orbital operates the hosted registry service.
- Reference reactor contracts (Soroban Rust) — open for anyone to fork
- Multi-region reference architecture docs
- 2–3 additional SEPs authored

CLOSED CLOUD SCALE + RAILS BEGINS

- **Cloud:** multi-region deployment (3+ regions), 99.95% SLA
- Cloud: observability pipeline, anomaly detection, status page
- Cloud: integration connectors (Slack, Discord, PagerDuty, Datadog, Zapier)
- Cloud: replay UI, dashboard v2, team accounts, SSO for business tier
- Cloud: ABI Registry hosted service (verification, moderation)
- **Rails MVP begins:**
 - Immutable audit log service
 - Dedicated-tenancy orchestration
 - SOC 2 Type I → Type II
 - First compliance export adapters (FATF, OFAC screening)
 - Priority-fanout auction operator (off-chain matching engine)
- Orbital-operated Stellar nodes in 2+ regions — data sovereignty begins

- Series A: \$3M–\$8M at Year 3
- First flagship anchor customer (Year 3–4): \$100K–\$300K ARR
- First fintech platform customer (Year 4): \$200K–\$500K ARR
- Cloud customer count: 500–2,000 paying accounts

SUCCESS CRITERIA — YEAR 5

≥5,000

NPM DOWNLOADS / WEEK

\$2M–\$5M

ARR (70% CLOUD, 30% RAILS)

≥100

PUBLIC PRODUCTION APPS

ABI Registry

CANONICAL STELLAR STANDARD

≥3 SEPs

AUTHORED AND ACCEPTED

SOC 2 Type II

CERTIFIED; 2+ OP REGIONS

ERA III · YEARS 5–10

Trust & Settlement Layer

Be the trust layer for regulated money. Capture the AI-agent economy.

Mission: become the trust layer for regulated money on Stellar; capture the AI-agent economy; monetize data sovereignty.

OPEN SOURCE PROTOCOL EXPANSION

- `@orbital/x402` — Express/Next.js middleware for agent payments
- `@orbital/agent-sdk` — payment client for autonomous agents
- **Intent compiler** — natural-language/DSL → subscription-graph compiler. Open-sourced at maturity (Year 6–7).
- Graph runtime (executes compiled DAGs)
- ZK-proof generation library (Noir/RiscZero circuits) — OSS, runnable locally
- Verifier contracts on Stellar
- **Shadow-Fork simulator** — fork any ledger, inject hypothetical ops, replay Soroban invocations. OSS core; Orbital operates a hosted SaaS version.
- `@orbital/anchor-sdk` client (SEP-24/31 lifecycle events)
- 3–5 additional SEPs (reactor spec, intent schema, attestation format)

CLOSED RAILS SCALES + DATA MOAT

- **ZK attestation service at scale** — proof production with managed keys, TEEs, enterprise API. Every Rails webhook carries a verifiable proof.
- **Compliance export engine** — FATF travel rule, OFAC, SAR, regulator formatters
- **Event markets** — priority-tier access to real-time streams (latency premiums auctioned)
- **Parametric event-delivery insurance** — missed-webhook payouts from a collateralized pool
- **Hosted intent-compilation service** (LLM at scale, caching, multi-tenant billing)
- **Hosted Shadow-Fork** (pre-trade TCA, MEV analysis, audit simulation)
- Dedicated deployments (EU, India, customer-specific regions)
- Archival data API — the proprietary dataset built on years of operations

BUSINESS GEOGRAPHIC + ECOSYSTEM

- Orbital-operated nodes in Lagos, Nairobi, Mumbai, São Paulo by Year 7
- First CBDC pilot (Year 8–9): \$500K–\$2M/yr
- 10+ enterprise anchor customers: \$2K–\$10K/mo each

- AI-agent operator partnerships: 1,000+ integrations via @orbital/x402
- Series B: \$20M–\$50M at Year 6–7
- Rails overtakes Cloud in revenue (Year 7–8)
- First \$1M ARR single customer (Year 9)

SUCCESS CRITERIA — YEAR 10

\$15M–\$30M ARR · 60% RAILS / 30% CLOUD / 10% DATA	≥1,000 X402 AI-AGENT INTEGRATIONS
≥5 regions ORBITAL-OPERATED NODES	≥5 SEPs AUTHORED
≥10 ENTERPRISE ANCHOR CUSTOMERS	1st CBDC PARTNERSHIP SIGNED · 30–50 FTE

Protocol Permanence & Accreditation

Be the layer that *is* Stellar to the outside world.

Mission: become the layer that is Stellar to the world outside Stellar. The layer regulators, central banks, and enterprise procurement assume exists.

OPEN SOURCE STANDARDS-SETTING

- **Orbital Network operator node software** — OSS, runnable by any certified third party
- Routing protocol specification (SEP-level standard)
- Identity layer reference implementation (passkey + federated Stellar addresses)
- Reactor-contract library expansion (hundreds of community-contributed patterns)
- 10+ SEPs authored or co-authored across identity, events, reactors, x402, compliance
- Intent-compiler v2 (multi-chain intent resolution reference)

CLOSED OPERATION + ACCREDITATION

- **The Orbital Network** — Cloudflare-style network of certified third-party operators. Orbital sets the protocol, certifies operators, maintains client-facing API. Removes single-point-of-failure objection from enterprise procurement.
- **Accreditation / oracle business** — Orbital rates, insures, and certifies other Stellar infrastructure (anchors, contracts, relayers). "Moody's for Stellar." Revenue: subscription + per-rating fees.
- **Identity standard (operated)** — `@orbital/auth` matures into Stellar's "Sign in with Google." Orbital operates the federated identity service.
- **CBDC infrastructure partnerships** — 3–5 central banks running on Orbital Rails. Bespoke closed integrations.
- **AI-agent compliance infrastructure** — regulator-facing surveillance APIs for autonomous agent activity. First-mover on a category regulators will mandate by Year 15–18.

BUSINESS COMMERCIAL MILESTONES

- Series C or late-stage raise: \$50M–\$150M at Year 12–14
- Strategic acquisition offers begin (Stripe, Circle, Visa, Mastercard, large bank)
- Exit optionality: SDF acquisition, payments conglomerate, or IPO at \$2B–\$5B valuation

SUCCESS CRITERIA — YEAR 20

\$75M–\$250M

ARR

≥50 operators

ORBITAL NETWORK ACROSS ≥15 REGIONS

≥80%

MAJOR STELLAR APPS USE ORBITAL IDENTITY

≥3 CBDCs

INFRASTRUCTURE PARTNERSHIPS

AI-agent compliance

REGULATORY DEFAULT

≥10 SEPs

AUTHORED · \$10M+ ACCREDITATION ARR

Seven Operating Principles

1 — PROTOCOL FIRST, OPERATIONS SECOND

Every feature ships as an OSS SDK + reference implementation first. The hosted operation follows. Never close what should be open.

2 — REGULATED MONEY IS THE NORTH STAR

Every product decision is scored against "does this help us win a fintech / anchor / CBDC?" — not against developer dopamine.

3 — SEP AUTHORSHIP OVER FEATURE SHIPPING

A feature shipped is a feature. A standard written is a decade of protocol-level leverage. When forced to choose, write the standard.

4 — RELIABILITY IS A BALANCE-SHEET ASSET

A missed webhook is a lost enterprise customer. Operational excellence is not a department — it is the product.

5 — DEVELOPER TRUST COMPOUNDS

A developer who uses Orbital for events tries it for hooks, then auth, then x402. Every positive interaction is a long-term balance-sheet entry.

6 — OSS NEAR-PARITY, CLOSED OPERATIONS

The OSS can run production workloads. The moat is not code — it is certifications, uptime history, ZK keys, regulator relationships, and the hosted operation of everything we open-source.

7 — ARCHITECT FOR THE NEXT ERA NOW

Postgres in Era I. Multi-region in Era II. ZK in Era III. Decentralized operators in Era IV. Each bet is placed at the right time or it is never placed at all.

Condensed 20-Year Roadmap

Period	Phase	OSS deliverable	Closed deliverable	Revenue target
Months 1–6	Foundation	v1.0 engine, full taxonomy, Postgres, HA	Billing, dashboard, API keys	—
Months 6–12	Grant capture	React hooks v1, ABI client, first SEP draft	Cloud single-region, observability	\$0–\$10K MRR
Year 1–2	Ecosystem planting	Full SDK suite reference, tutorials, boilerplates	Cloud multi-region, integrations	\$100K–\$300K ARR
Year 2–3	Managed SaaS scale	@orbital/payments , @orbital/auth OSS	SOC 2 Type II, Rails MVP	\$500K–\$2M ARR
Year 3–5	Runtime ownership	ABI Registry standard, reactor contracts, 3+ SEPs	Rails scales, first anchor customers	\$2M–\$5M ARR
Year 5–7	Trust layer	x402, agent-SDK, ZK proof library, Shadow-Fork OSS	ZK attestation service, compliance exports, event markets	\$5M–\$15M ARR
Year 7–10	Data sovereignty	Intent compiler OSS, anchor-SDK	Archival data API, priority-fanout, parametric insurance, first CBDC	\$15M–\$30M ARR
Year 10–15	Network decentralization	Orbital Network operator software, identity standard, 10+ SEPs	Orbital Network certification, identity service, 2+ CBDCs	\$30M–\$100M ARR
Year 15–20	Protocol permanence	Reactor library ecosystem, multi-chain intent reference	Accreditation business, AI-agent compliance, 3+ CBDCs	\$100M–\$250M ARR

Three Strategic Calls for This Quarter

1 — Commit to the regulated-money north star

Internal memo; all hiring, pitching, and pricing after this date flows from it. Dev-tool motion remains active but as top-of-funnel, not the thesis.

2 — Publish the license and contribution policy

`LICENSE` = MIT for all OSS packages. `CONTRIBUTING.md` written. Stability pledge for v1.0.0 documented. Decide once, move on.

3 — Write the first SEP

Choose one: standardized Soroban event schema, reactor contract spec, or webhook delivery semantics. Target: submitted to SDF before Month 6. This is the single highest-leverage activity in Era I.

Final Word

Twenty years is not a schedule — it is a direction. The specific products listed above will shift. Competitors will rise and fall. x402, ZK, CBDCs, AI agents are bets, not certainties.

What is not a bet: **Stellar will grow as a settlement layer for real-world money**, and **everything that grows on it will need to emit and consume events**. The company that owns the protocol layer between Stellar and the rest of the financial world owns the economic surface of Stellar's future.

Orbital does not need to be the biggest blockchain company. Orbital needs to be **indispensable to every serious entity moving money on Stellar**. Indispensability at the protocol layer, in a network growing toward mainstream financial adoption, is one of the most durable business positions available in technology.

Build the protocol correctly. Operate the hosting reliably. Win trust one integration at a time. The twenty years take care of themselves.

